

Introduction

Aiming

- know the effect of static loading and dynamic loading
- in aircraft structure, phenomena such as flutter may cause structural failure

Case study

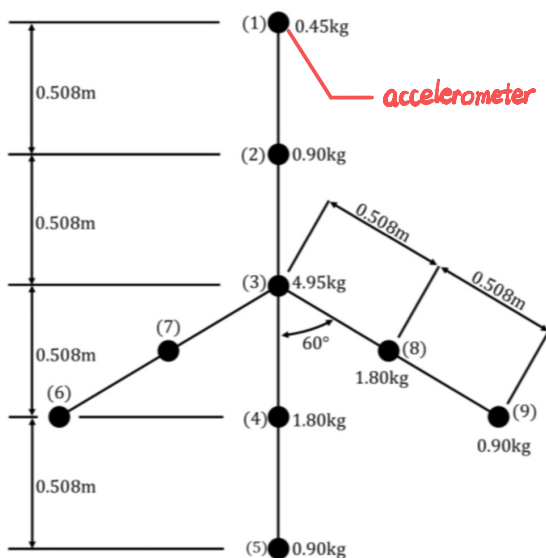
German V-2 rockets suffered structure failure due to flutter on the top plane structure during WWII

Procedure

Objective

- measure undamped nature frequency $\omega_n = \sqrt{\frac{F}{R}}$
- measure model damping ratio $\xi = \frac{c}{2\sqrt{km}}$
- measure first mode shape Φ

Apparatus



Procedure

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① Measure the response at the accelerometer & record f_{applied} (Hz), V , $\ddot{r}_0$ (m/s²), θ_0 (rad).

② Measure deflection at w_u record r (m).

Theory

w_u and ξ

$$m\ddot{r} + c\dot{r} + kr = 0 \xrightarrow{r=r_0 e^{pt}} mp^2 + cp + k = 0 \longrightarrow p = \frac{c \pm \sqrt{c^2 - 4km}}{2m}$$

$$\text{for } c^2 - 4km < 0, p = \frac{c}{2m} \pm i \frac{\sqrt{4km - c^2}}{m} = \alpha + i\beta \longrightarrow r = r_0 e^{\alpha} e^{i\beta} + r_0 e^{\alpha} e^{-i\beta}$$

$$\text{meanwhile, } \ddot{r} + \frac{c}{m}\dot{r} + \frac{k}{m}r = 0 = \ddot{r} + (2w_u \xi)\dot{r} + w_u^2 r = 0 \quad \therefore w_u = \sqrt{\frac{k}{m}}, \quad \xi = \frac{c}{2\sqrt{km}}$$

$$m\ddot{r} + c\dot{r} + kr = R$$

$$R = R_0 e^{i\omega t} \longrightarrow r = r_0 e^{i\omega t} \text{ OR } r_p \cos(\omega t) - r_a \sin(\omega t) \text{ OR } r_0 \cos(\omega t + \theta)$$

$$\therefore (-\omega^2 m + i\omega c + k) r_0 = R_0$$

$$\therefore r_0 = \frac{1}{-\omega^2 m + i\omega c + k} R_0 = \frac{k - \omega^2 m - i\omega c}{(k - \omega^2 m)^2 - (\omega c)^2} R_0$$

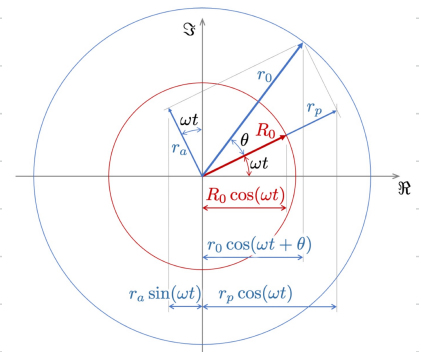
$$\therefore r = \frac{k - \omega^2 m}{(k - \omega^2 m)^2 - (\omega c)^2} R_0 \cos(\omega t) + \frac{\omega c}{(k - \omega^2 m)^2 - (\omega c)^2} R_0 \sin(\omega t)$$

$$\text{Also, for } r = r_0 \cos(\omega t + \theta)$$

$$\dot{r} = -\omega r_0 \sin(\omega t + \theta) = \omega r_0 \cos(\omega t + \theta + \frac{\pi}{2}) = v_0 \cos(\omega t + \theta_v)$$

$$\ddot{r} = -\omega^2 r_0 \cos(\omega t + \theta) = \omega^2 r_0 \cos(\omega t + \theta) = a_0 \cos(\omega t + \theta_a)$$

displacement is out-of-phase with the load and delayed with the respect to the load

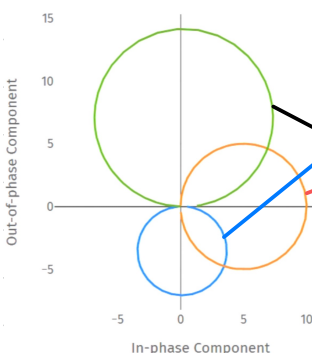


$$r_p^2 + (r_a^2 + \frac{R_0^2}{2c^2 \omega^2})^2 = (r_0 \cos \theta)^2 + (r_0 \sin \theta + \frac{R_0}{2c \omega})^2 = \frac{1}{\omega^2} (\frac{R_0}{2c})^2$$

$$(v_p - \frac{R_0}{2c})^2 + v_a^2 = (v_0 \cos \theta - \frac{R_0}{2c})^2 + (v_0 \sin \theta)^2 = (\frac{R_0}{2c})^2$$

$$a_p^2 + (a_a^2 - \frac{R_0 \omega}{2c})^2 = (a_0 \cos \theta)^2 + (a_0 \sin \theta - \frac{R_0 \omega}{2c})^2 = \omega^2 (\frac{R_0}{2c})^2$$

* when $\omega \rightarrow w_u$ for very small c , r and a form circle



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when $\frac{d||\dot{r}||}{d\omega} = 0 \longrightarrow \dot{r}$ is maximum for $\omega = \omega_u$

when $\gamma_a = 0$ and $\tan\theta_v = 0 \longrightarrow$ for $\omega = \omega_u$

Obtaining ω_u and ξ from data using pairs of data $\omega_{A^*}, \theta_{vA}$ and $\omega_{B^*}, \theta_{vB}$

$$\cdot \omega_u^2 = \frac{\omega_{A^*} \omega_{B^*} \tan\theta_{vB} - \omega_{B^*} \omega_{A^*} \tan\theta_{vA}}{\omega_{B^*} \tan\theta_{vB} - \omega_{A^*} \tan\theta_{vA}}$$

$$\cdot \frac{c}{m} = \frac{\omega_{B^*}^2 - \omega_{A^*}^2}{\omega_{A^*} \tan\theta_{vA} - \omega_{B^*} \tan\theta_{vB}}$$

$$\cdot \xi = \frac{1}{2\omega_n} \frac{c}{m}$$

Observation



will affect the results as pendulum



vibrate less when close to ω_n

point mass 6 and 9 at wing tip vibrate strongly

other point mass especially for fuselage hard to observe



many wire attached

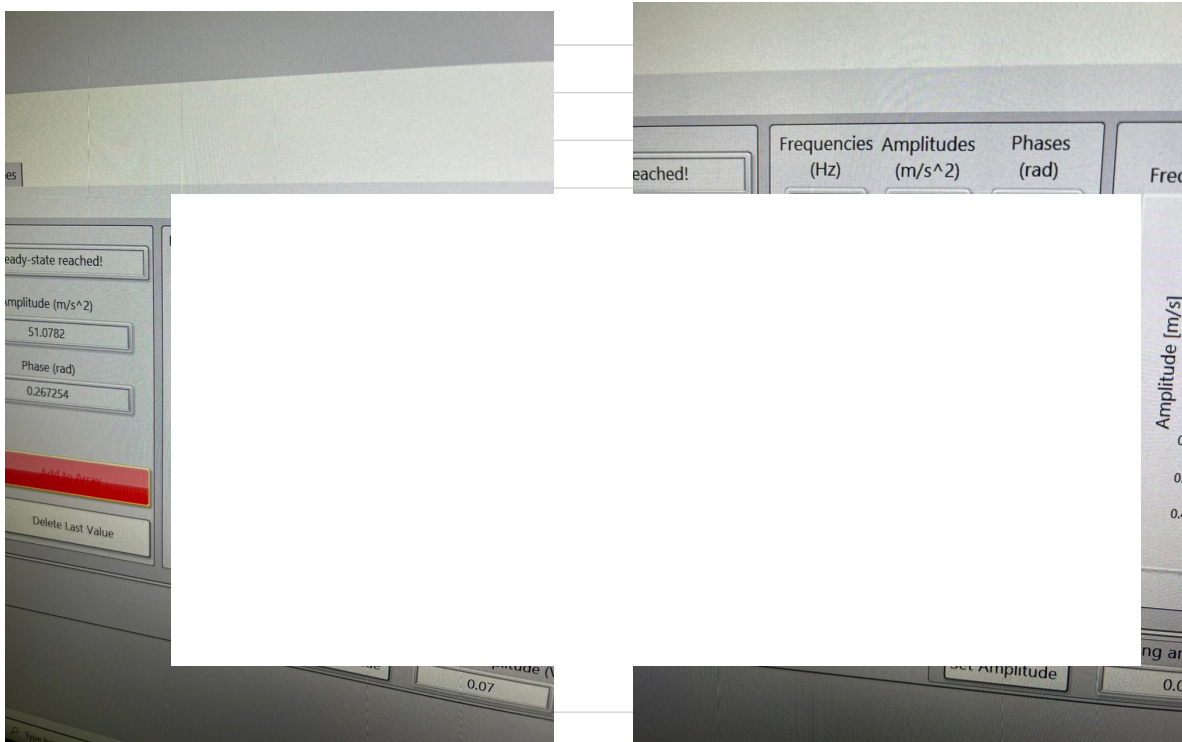
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Data Recording

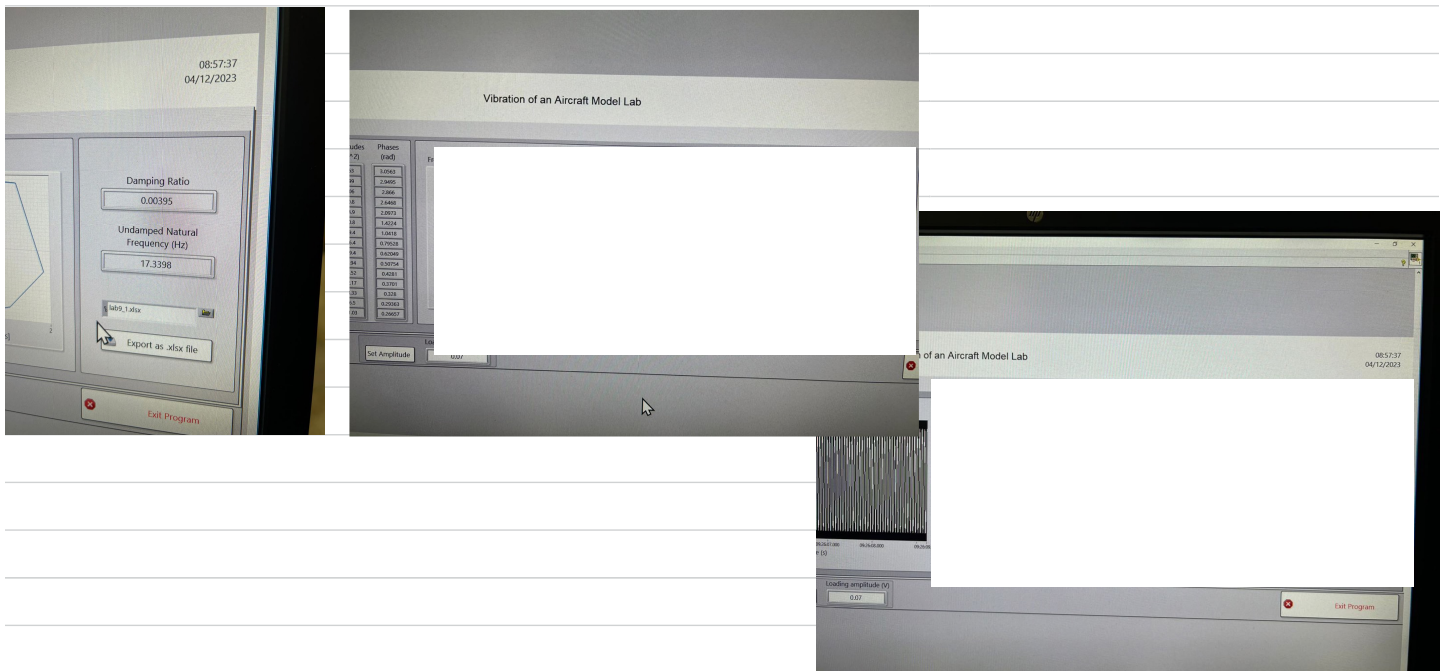
For change ω at accelerometer 9

doing experiment for 2-times to get more accurate results

however, two sets of data at lapping applied frequency shows different a .

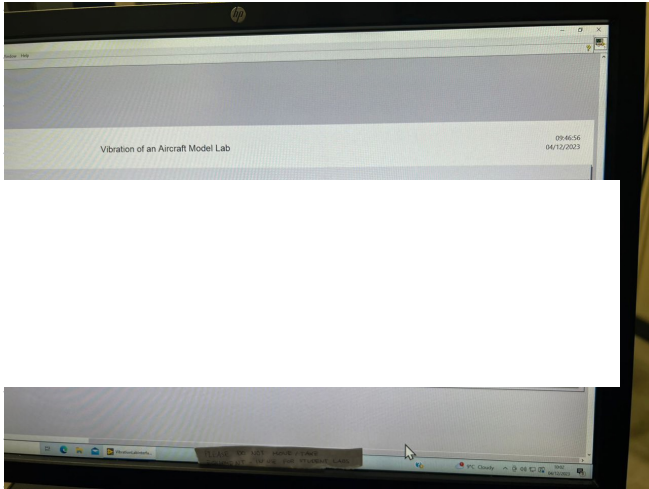


Another plots for test 1

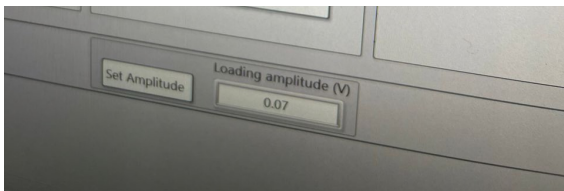
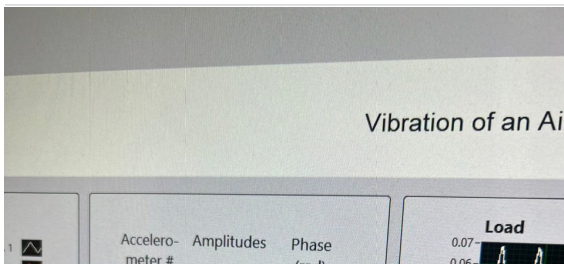


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Another plots for test 2



Carry on test 2 which gives much more accurate & approximation



Data Analysis

For accelerometer 9

	Applied		Read		Adjustment	Calculation									
	Frequency ω (Hz)	Loading amplification factor V	Acceleration amplitude a_0 (m/s^2)	Phase θ_s (rad)	Frequency ω (rad/s)	a_p (m/s^2)	a_s (m/s^2)	Velocity amplitude v_0 (m/s)	θ_v (rad)	v_p (m/s)	v_s (m/s)	Displacement amplitude r_0 (m)	θ (rad)	r_p (m)	r_s (m)
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2															
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5															
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Calculate ω_n and ξ

	Damping ratio ξ	ω^2 (rad^2/s^2)	Frequency ω (Hz)	Frequency ω (rad/s)					
Read for test 1									
Read for test 2									
2 Data Points									
4 Data Points									
6 Data Points									
8 Data Points									

Sampled Data						Calculation		
Applied	Calculation							
Frequency ω (Hz)	Frequency ω (rad/s)	θ_v (rad)	$\tan(\theta_v)$			ω^2 (rad^2/s^2)	c/m	
9					9 & 10			
10					10 & 11			
11					11 & 12			
12					12 & 13			
13					13 & 14			
14					14 & 15			
15					15 & 16			
16					9 & 11			
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					9 & 16			

Find mode shape Φ_i

	Point mass	Applied		Read		Normalisation
		mass m (kg)	Loading amplification factor V	Amplitude r_0 (m)	Phase θ_a (rad)	Eigenvectors
Fuselage	1					
	2					
	3					
	4					
	5					
Wing	6					
	7					
	8					
	9					

Figure 8: Amplitude of a v.s. frequency

Figure 9: Amplitude of v v.s. frequency

Figure 10: Amplitude of r v.s. frequency



Figure 11: Amplitude v.s. Frequency

π



$-\pi$

Figure 15: Phase v.s. Frequency

Figure 122: Nyquist plot for a

Figure 13: Nyquist plot for v

Figure 14: Nyquist plot for r

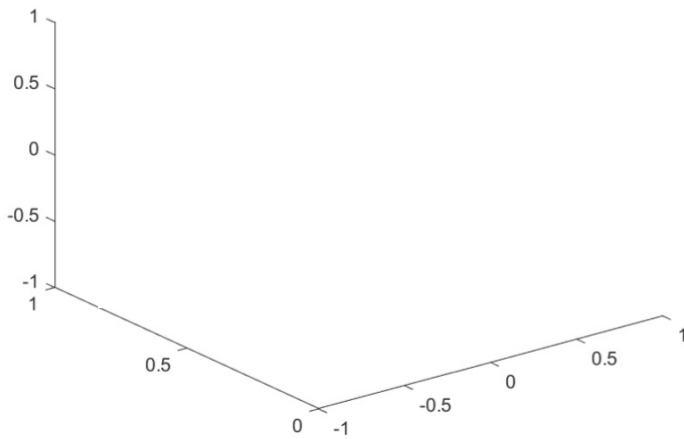


Figure 16: mode shape in experiment

effect by string

Figure 17: experiment mode shape v.s. FEA mode shape for fuselage

Figure 18: experiment mode shape v.s. FEA mode shape for wing

Error Analysis

Source of error

- imperfect in plane model
- plane model not being placed perfectly symmetric
- accelerometers not exactly in stated positions
- electrical noises $\leftarrow$ increase V to reduce its proportion in data to reduce percentage error

String length

length $\uparrow \rightarrow F$ due to gravity $\downarrow \leftarrow$ increase L to reduce error